

The empirical recurrence for OEIS A283689: recovering a conjecture that is not written down

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Abstract

OEIS A283689 records “Empirical recurrence of order 83”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 116$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 83, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 83 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A283689 is “Number of nX6 0..1 arrays with no 1 equal to more than two of its horizontal, diagonal and antidiagonal neighbors.”. It has offset 1 and begins

64, 1824, 47829, 1392868, 39372084, 1119189256, 31815868749, 904083654190, ...

The entry states:

Empirical recurrence of order 83 (see link above)

As of the “Last modified” line on the live entry (Mar 14 2017 16:15:03, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1888$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 116$ of the 1888, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 116$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 232$ exact terms of the model returns order 83 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{83} c_i a(n-i),$$

c_1	20	c_{22}	−13305273255030	c_{43}	−12471511919526082	c_{64}	−176536681073072
c_2	191	c_{23}	−45336856191326	c_{44}	−19157032115708072	c_{65}	70529277807696
c_3	1437	c_{24}	−71352694335855	c_{45}	3740883166164348	c_{66}	5928671185102
c_4	−945	c_{25}	−16155999842619	c_{46}	5500709762436506	c_{67}	38025246230389
c_5	−5317	c_{26}	188339745249560	c_{47}	21880777212836895	c_{68}	−15427203886385
c_6	−487705	c_{27}	430749628190138	c_{48}	4755877216655521	c_{69}	−807256205628
c_7	−2014854	c_{28}	208345506956346	c_{49}	4693074700692129	c_{70}	−2469683746469
c_8	−6349346	c_{29}	−621354480295036	c_{50}	−12895668476367548	c_{71}	1047342902120
c_9	36302478	c_{30}	−1722014970188431	c_{51}	−6344025523090717	c_{72}	20366240368
c_{10}	303531349	c_{31}	−1467844175606869	c_{52}	−6116276064886878	c_{73}	123197591785
c_{11}	1100299856	c_{32}	544598113799033	c_{53}	1199797271627025	c_{74}	−60337236890
c_{12}	418680178	c_{33}	3104843597836655	c_{54}	5487261343284015	c_{75}	5711486460
c_{13}	−10188048476	c_{34}	4452955420038919	c_{55}	1948645443919631	c_{76}	−970930678
c_{14}	−50141356034	c_{35}	694726109421607	c_{56}	2009022599791858	c_{77}	979709185
c_{15}	−105227025604	c_{36}	−2652387706166623	c_{57}	−3446888091985601	c_{78}	506201158
c_{16}	−18862947109	c_{37}	−6513056527999481	c_{58}	−324324921962122	c_{79}	−220454055
c_{17}	581184828935	c_{38}	945648851898383	c_{59}	−1130968588284326	c_{80}	28886337
c_{18}	2045789505005	c_{39}	7239864253215107	c_{60}	1303442227247806	c_{81}	−15474044
c_{19}	3812599501674	c_{40}	10758691277515236	c_{61}	404225359657121	c_{82}	2033696
c_{20}	4681655972950	c_{41}	4874755621111627	c_{62}	17378827458940	c_{83}	329216
c_{21}	767794245291	c_{42}	−10302111636125454	c_{63}	−202245555449102		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 83, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $83 + 4$ terms removed returns order 83 as well, so $\deg q_{\min} = 83 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $83 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 83 that this sequence satisfies — and that family turns out to have exactly one member.

References

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